Meta-analysis technical report

Evolution of ISUP Grade Group 1 Detection Across Prostate Cancer Screening Strategies

Analysis: meta_analysis.R · Numeric log: output/results.txt · Source: primary-outcome table

R 4.6.0; `meta` 8.5.0; `metafor` 5.0.1.

1. Data and analysis units

The extraction contains nine randomized trials and 18 arms. ProScreen (two arms) is excluded from every analysis, leaving eight trials and 16 arms. Comparative models use paired randomized arms; proportion models use 14 study-strategy units.

Cohort	Included trials	Use
Screening invitation vs no organised screening	ERSPC, PLCO, CAP	Pooled comparative RRs
Diagnostic/biopsy strategy; both arms screened	STHLM3-MRI, MVP, GÖTEBORG-2, PROSA	Pooled comparative RRs
Timing of screening initiation	PROBASE	Standalone estimate

GÖTEBORG-2 arms assigned to the same broad `PSA → MRI` category are combined. Different strategies within a trial remain separate. PROBASE contributes only its immediate PSA arm to strategy proportions. Its delayed comparator included 6,537 DRE assessments and two cancers, but no coherent PSA-screened denominator at this endpoint.

CAP randomized 573 practices. No ICC or cluster-adjusted grade-specific variance was available, so its confidence intervals and weights are unadjusted; all invitation-family outcomes are also analyzed without CAP.

Grade-yield inputs

`NR` denotes no coherent reported denominator.

Trial	GG1 / biopsied, I	GG1 / biopsied, C	GG≥2 / biopsied, I	GG≥2 / biopsied, C
ERSPC	4,325 / 17,543 (24.7%)	NR	1,665 / 17,543 (9.5%)	NR
PLCO	NR	NR	NR	NR
CAP	NR	NR	NR	NR
PROBASE	15 / 120 (12.5%)	2 / 37 (5.4%)	33 / 120 (27.5%)	0 / 37 (0.0%)
STHLM3-MRI	34 / 311 (10.9%)	73 / 438 (16.7%)	192 / 331 (58.0%)	106 / 438 (24.2%)
MVP	4 / 24 (16.7%)	4 / 28 (14.3%)	11 / 24 (45.8%)	4 / 28 (14.3%)
GÖTEBORG-2	68 / 254 (26.8%)	159 / 588 (27.0%)	117 / 254 (46.1%)	139 / 588 (23.6%)
PROSA	9 / 40 (22.5%)	4 / 20 (20.0%)	17 / 40 (42.5%)	7 / 20 (35.0%)

Trial	GG≥2 / all detected cancers, I	GG≥2 / all detected cancers, C
ERSPC	1,665 / 5,990 (27.8%)	1,947 / 4,307 (45.2%)
PLCO	1,104 / 3,452 (32.0%)	1,120 / 2,974 (37.7%)
CAP	6,649 / 12,013 (55.3%)	8,019 / 12,958 (61.9%)
PROBASE	33 / 48 (68.8%)	0 / 2 (0.0%)
STHLM3-MRI	192 / 226 (85.0%)	106 / 179 (59.2%)
MVP	11 / 15 (73.3%)	4 / 8 (50.0%)
GÖTEBORG-2	117 / 185 (63.2%)	139 / 298 (46.6%)
PROSA	17 / 26 (65.4%)	7 / 11 (63.6%)

CAP is excluded from grade-per-biopsy analyses because its 5,848 biopsies concern the first screening round, while grade counts are cumulative over 15 years. STHLM3-MRI reports intervention GG≥2 as `192/331`, although its general biopsy count is 311; the outcome-specific denominator is retained.

2. Analysis specification

- Detection-rate RRs use randomized/invited denominators, preserving the randomized comparison.
- Detection-rate proportions are reported on both denominators: per man screened (protocol s.9) and per man randomized/invited.
- Cancer-grade composition uses all detected prostate cancers.
- Biopsy yield uses outcome-specific biopsy denominators.
- Biopsy burden is reported both comparatively (RR of biopsies per randomized/invited participant, trials reporting both arms) and per screened/assessed participant.
- Invitation and diagnostic-strategy cohorts are fitted separately.
- Strategy proportions are fitted separately and are not head-to-head comparisons.

Per-screened detection rates carry a denominator caveat: in ERSPC, PLCO and CAP the numerators accrue across the whole randomized arm over long follow-up, while the per-screened denominator counts only men actually tested, so those rates are upward-biased. The per-randomized/invited rates are the internally consistent ones. Both are reported so the gap is visible rather than assumed.

Models:

- Risk ratios: inverse-variance random effects, DerSimonian–Laird variance, Hartung–Knapp confidence intervals, and a 0.5 all-cell correction for zero cells.
- Proportions: random-effects logit-normal GLMMs; exact binomial intervals for single-study groups.
- Single-trial contrasts are not pooled: they carry the trial's own estimate with a normal-approximation interval, and no heterogeneity statistics.
- Prediction intervals are reported for models with at least three studies.
- Leave-one-out analysis omits each contribution once, and runs only where at least three studies contribute; omitting one of two would leave a single trial rather than a pooled estimate.

3. Comparative risk ratios

3.1 GG1 detection per randomized/invited participant

Cohort	k	RR (95% CI)	I²	95% prediction interval
Invitation vs usual care	3	1.573 (0.726–3.408)	99.3%	0.335–7.378
Diagnostic/biopsy strategy	4	0.563 (0.152–2.081)	74.6%	0.093–3.397
PROBASE timing contrast	1	7.487 (1.712–32.736)	—	—

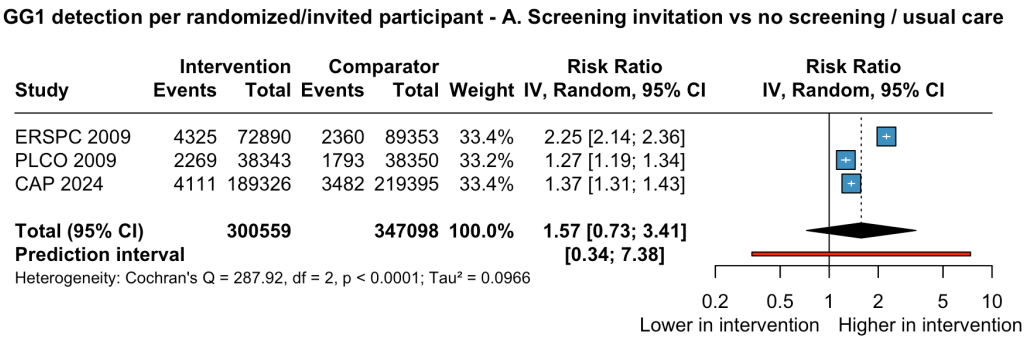


Figure 1a. Forest plot for GG1 detection per randomized/invited participant — screening invitation vs no organised screening.

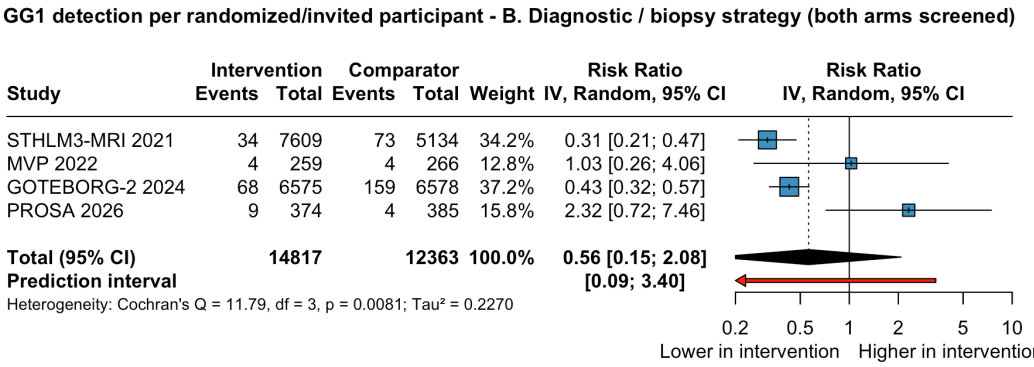


Figure 1b. Forest plot for GG1 detection per randomized/invited participant — diagnostic/biopsy strategy.

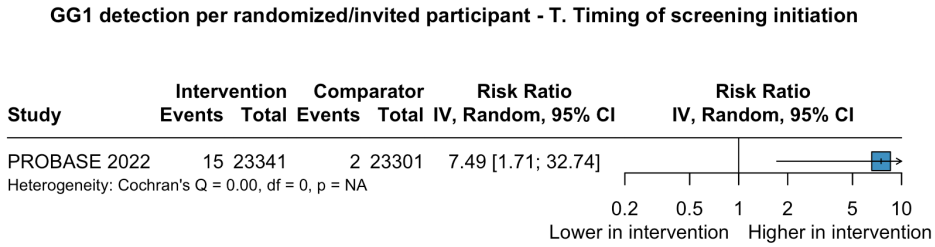


Figure 1c. Forest plot for GG1 detection per randomized/invited participant — PROBASE timing-of-screening contrast.

3.2 GG1 among all detected prostate cancers

Cohort	k	RR (95% CI)	I²	95% prediction interval
Invitation vs usual care	3	1.223 (0.952–1.572)	96.7%	0.750–1.997
Diagnostic/biopsy strategy	4	0.567 (0.302–1.064)	68.9%	0.152–2.113
PROBASE timing contrast	1	0.380 (0.198–0.729)	—	—

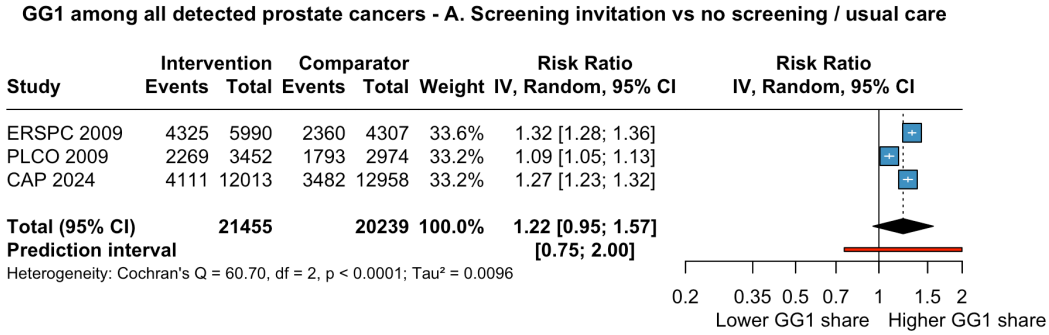


Figure 2a. Forest plot for GG1 among all detected prostate cancers — screening invitation vs no organised screening.

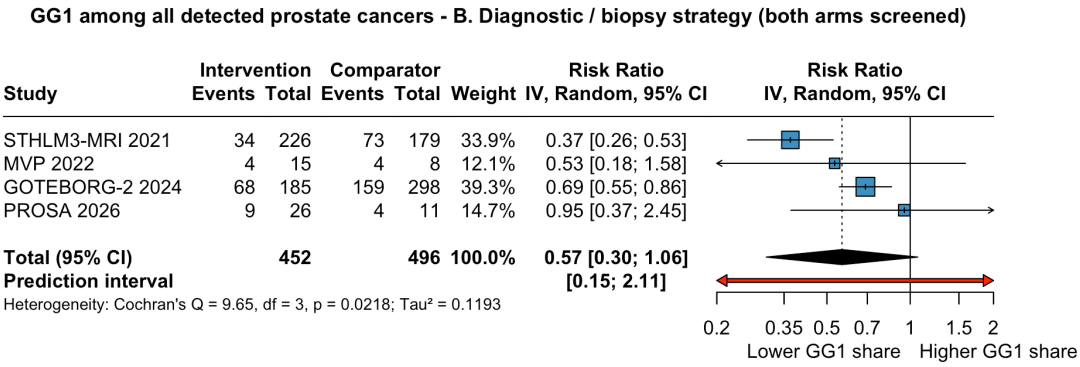


Figure 2b. Forest plot for GG1 among all detected prostate cancers — diagnostic/biopsy strategy.

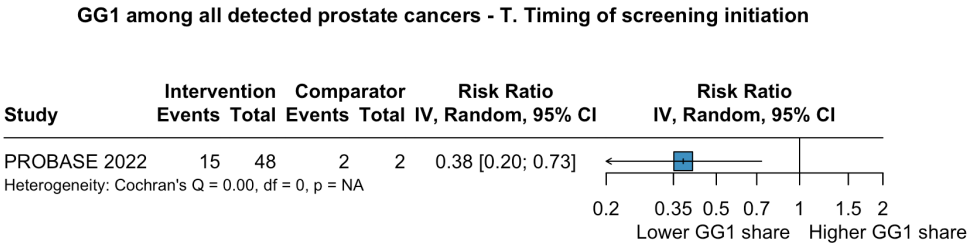


Figure 2c. Forest plot for GG1 among all detected prostate cancers — PROBASE timing-of-screening contrast.

3.3 GG≥2 detection per randomized/invited participant

Cohort	k	RR (95% CI)	I²	95% prediction interval
Invitation vs usual care	3	0.993 (0.886–1.113)	64.6%	0.805–1.225
Diagnostic/biopsy strategy	4	1.292 (0.571–2.925)	73.8%	0.373–4.475
PROBASE timing contrast	1	66.885 (4.098–1091.529)	—	—

The PROBASE comparator has zero inferred GG≥2 events; continuity-correction sensitivity is reported below.

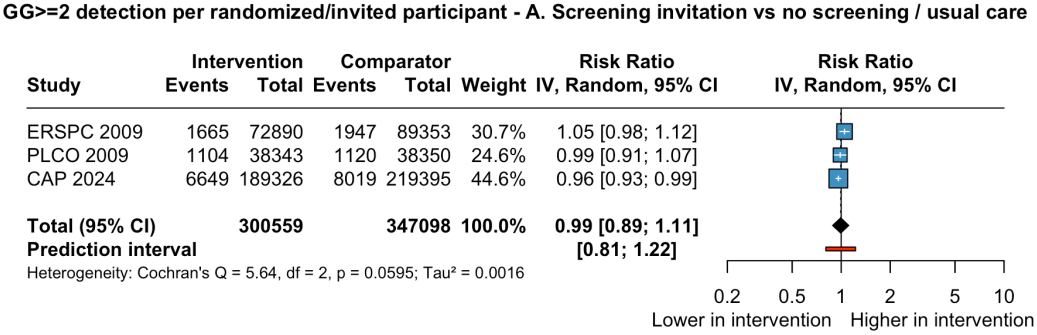


Figure 3a. Forest plot for GG≥2 detection per randomized/invited participant — screening invitation vs no organised screening.

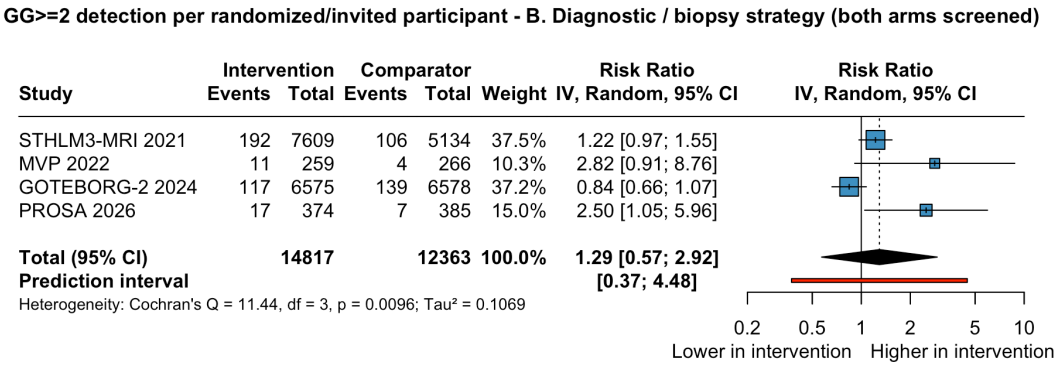


Figure 3b. Forest plot for GG≥2 detection per randomized/invited participant — diagnostic/biopsy strategy.

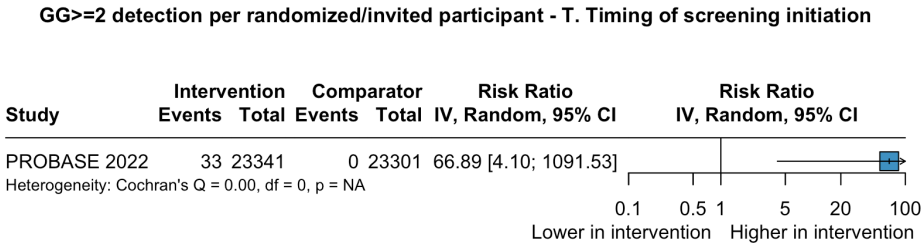


Figure 3c. Forest plot for GG≥2 detection per randomized/invited participant — PROBASE timing-of-screening contrast.

3.4 GG≥2 among all detected prostate cancers

Trial	Intervention strategy: GG≥2 / all cancers	Comparator strategy: GG≥2 / all cancers	Trial RR
STHLM3-MRI	Biomarker → MRI: 192/226 (85.0%)	PSA: 106/179 (59.2%)	1.435
MVP	MRI-first: 11/15 (73.3%)	PSA: 4/8 (50.0%)	1.467
GÖTEBORG-2	MRI-targeted only: 117/185 (63.2%)	Systematic ± targeted: 139/298 (46.6%)	1.356
PROSA	MRI-first: 17/26 (65.4%)	PSA-triggered MRI: 7/11 (63.6%)	1.027

For STHLM3-MRI, 85.0% / 59.2% gives RR 1.435. The pooled RR is 1.388, equivalent to a 38.8% higher GG≥2 share among detected cancers.

Cohort	k	RR (95% CI)	I²	95% prediction interval
Invitation vs usual care	3	0.776 (0.469–1.286)	98.8%	0.273–2.206
Diagnostic/biopsy strategy	4	1.388 (1.233–1.563)	0.0%	1.179–1.634
PROBASE timing contrast	1	4.102 (0.324–51.881)	—	—

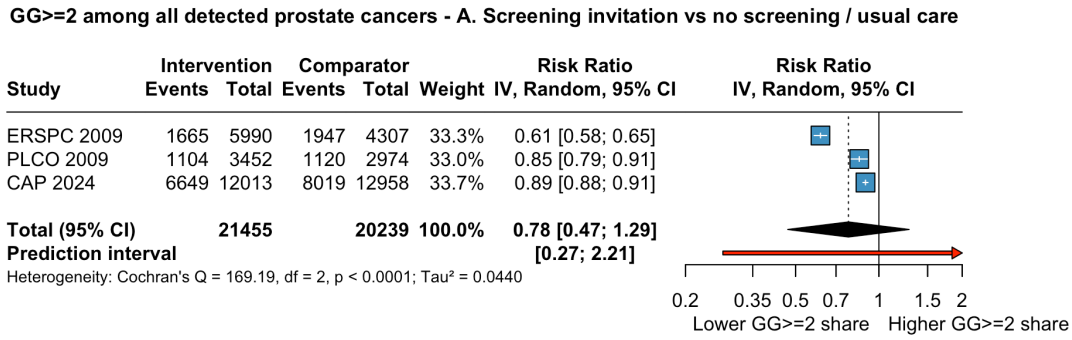


Figure 4a. Forest plot for GG≥2 among all detected prostate cancers — screening invitation vs no organised screening.

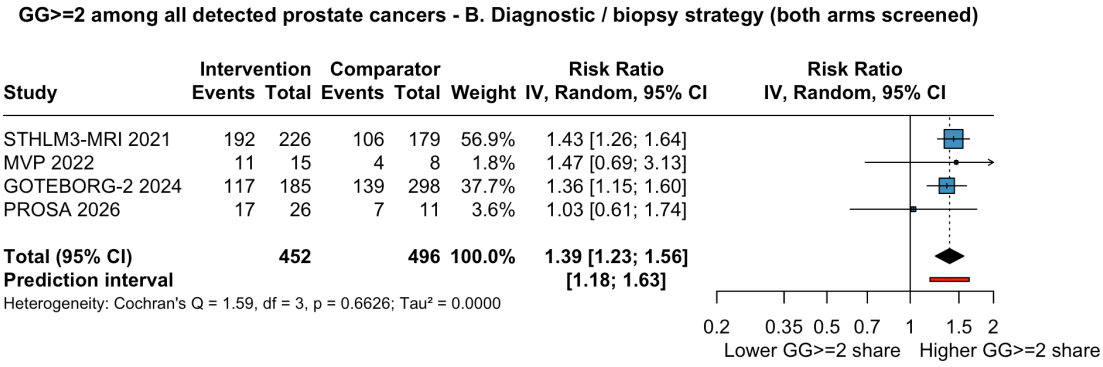


Figure 4b. Forest plot for GG≥2 among all detected prostate cancers — diagnostic/biopsy strategy.

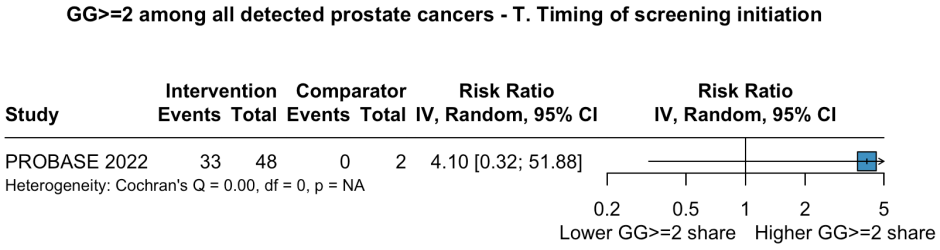


Figure 4c. Forest plot for GG≥2 among all detected prostate cancers — PROBASE timing-of-screening contrast.

3.5 GG1 among participants biopsied

Cohort	k	RR (95% CI)	I²	95% prediction interval
Diagnostic/biopsy strategy	3	1.002 (0.888–1.130)	0.0%	0.601–1.669
PROBASE timing contrast	1	2.313 (0.554–9.650)	—	—

STHLM3-MRI was excluded from the diagnostic/biopsy-strategy analysis.

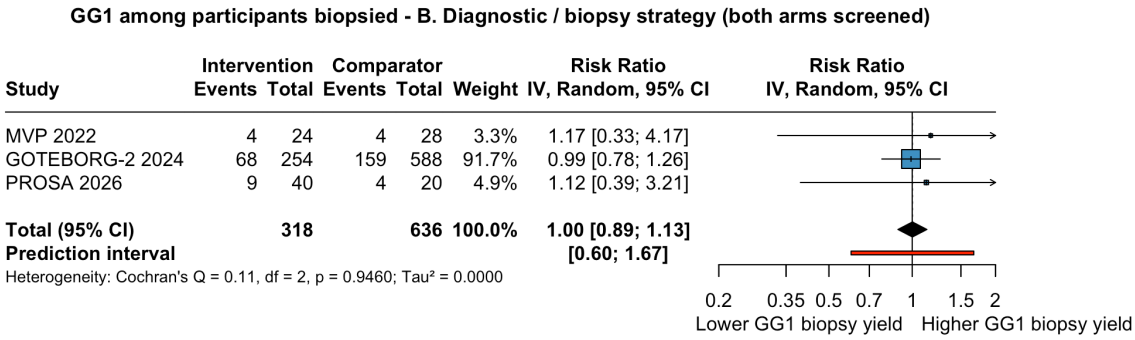


Figure 5a. Forest plot for GG1 among participants biopsied — diagnostic/biopsy strategy.

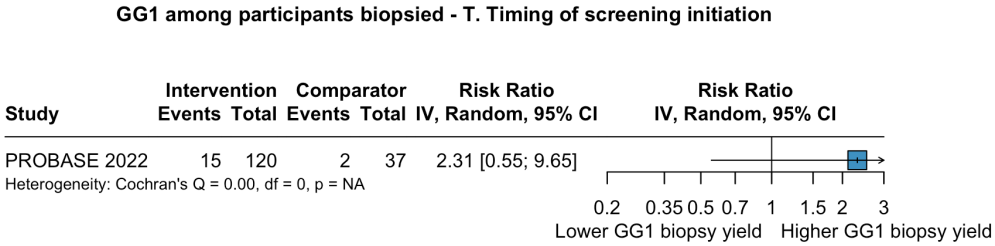


Figure 5b. Forest plot for GG1 among participants biopsied — PROBASE timing-of-screening contrast.

3.6 GG≥2 among participants biopsied

Cohort	k	RR (95% CI)	I²	95% prediction interval
Diagnostic/biopsy strategy	3	1.875 (0.861–4.085)	25.3%	0.646–5.442
PROBASE timing contrast	1	21.041 (1.320–335.297)	—	—

STHLM3-MRI was excluded from the diagnostic/biopsy-strategy analysis. No invitation-family biopsy-yield RR is estimable.

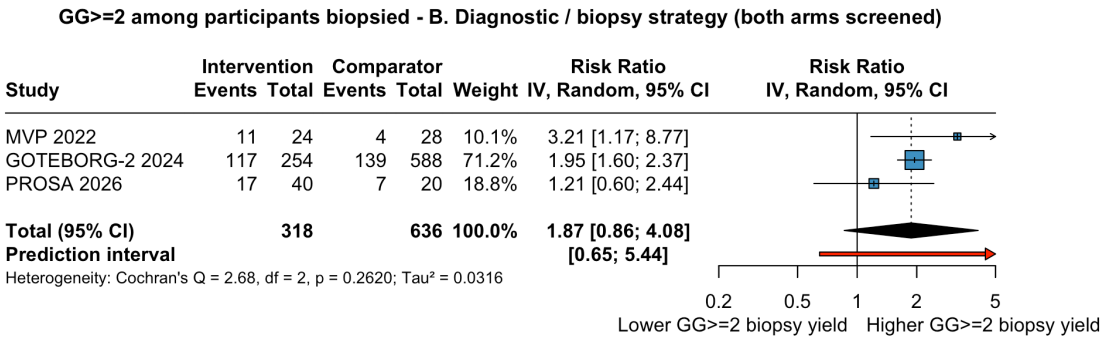


Figure 6a. Forest plot for GG≥2 among participants biopsied — diagnostic/biopsy strategy.

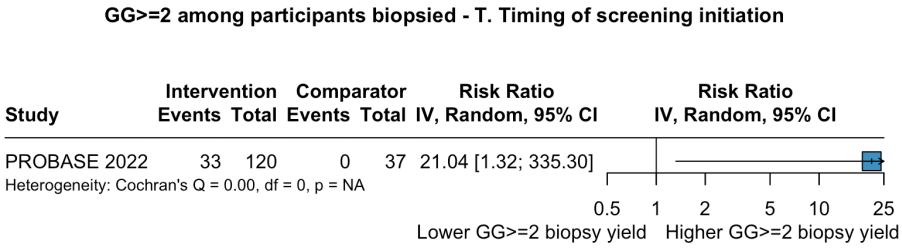


Figure 6b. Forest plot for GG≥2 among participants biopsied — PROBASE timing-of-screening contrast.

3.7 Biopsies performed per randomized/invited participant

Protocol secondary endpoint (s.6): the number of biopsies performed. This is the randomized comparison of biopsy burden, distinct from the yield outcomes above.

Cohort	k	RR (95% CI)	I²	95% prediction interval
Diagnostic/biopsy strategy	4	0.713 (0.236–2.152)	92.0%	0.175–2.900
PROBASE timing contrast	1	3.238 (2.241–4.678)	—	—

The pooled estimate is dominated by opposing trial effects: GÖTEBORG-2 (0.43) and STHLM3-MRI (0.48) roughly halve biopsies, while PROSA increases them (2.06) and MVP is neutral (0.88). With I² = 92.0% the pooled value should not be read as a single biopsy-burden effect. No invitation-family estimate is possible: ERSPC and CAP report intervention-arm biopsies only, and PLCO reports none.

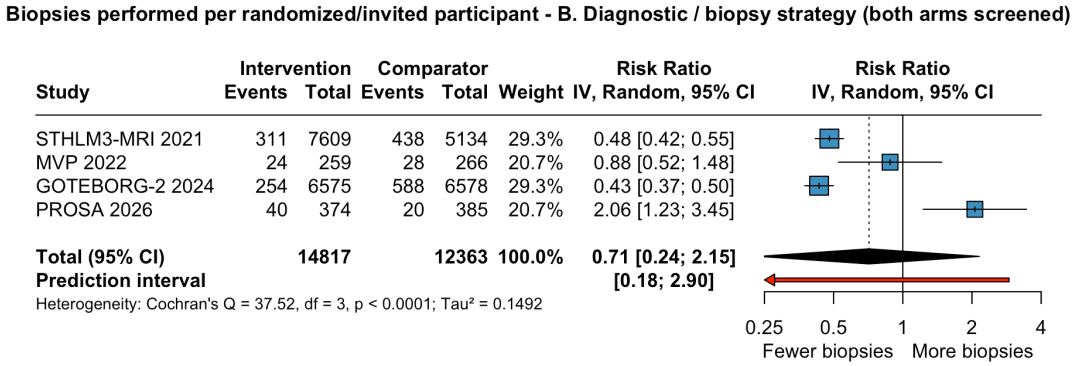


Figure 7a. Forest plot for Biopsies performed per randomized/invited participant — diagnostic/biopsy strategy.

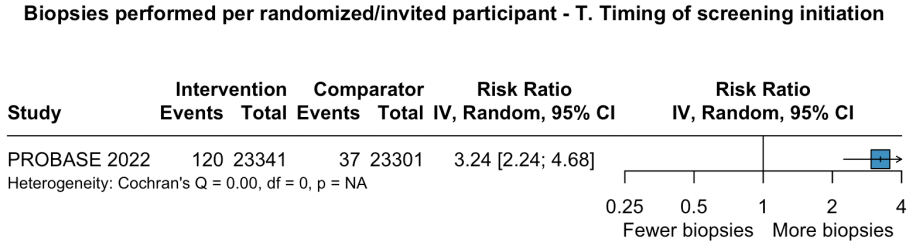


Figure 7b. Forest plot for Biopsies performed per randomized/invited participant — PROBASE timing-of-screening contrast.

4. Proportions by screening strategy

These are separate descriptive models, not randomized head-to-head comparisons. `k` is the number of study-strategy units.

4.1 GG1 per screened participant

Protocol-specified denominator (s.9): detection per number of screened men.

Strategy	k	Proportion (95% CI)	I²
No screening / usual care	3	2.702% (1.634–4.438)	99.9%
PSA + DRE	1	6.962% (6.688–7.244)	—
PSA	5	1.422% (0.317–6.140)	99.4%
PSA → MRI	2	1.707% (1.502–1.939)	0.8%
Biomarker → MRI	1	0.447% (0.310–0.624)	—
MRI-first	2	2.114% (1.231–3.606)	0.0%

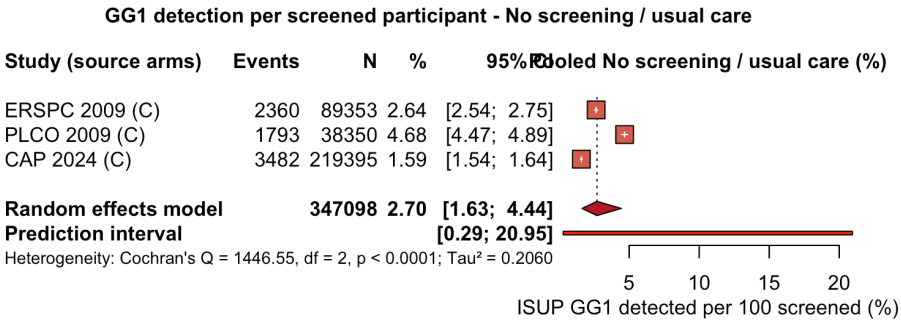


Figure 8a. Forest plot for GG1 per screened participant — no screening / usual care.

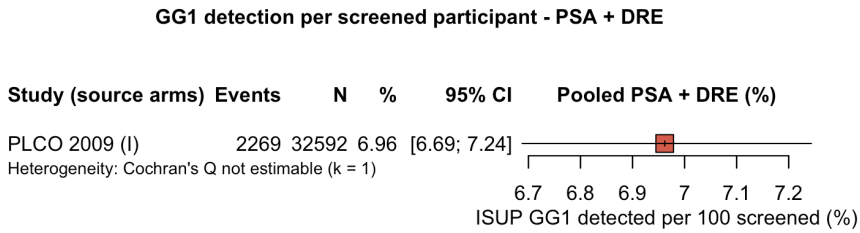


Figure 8b. Forest plot for GG1 per screened participant — PSA + DRE.

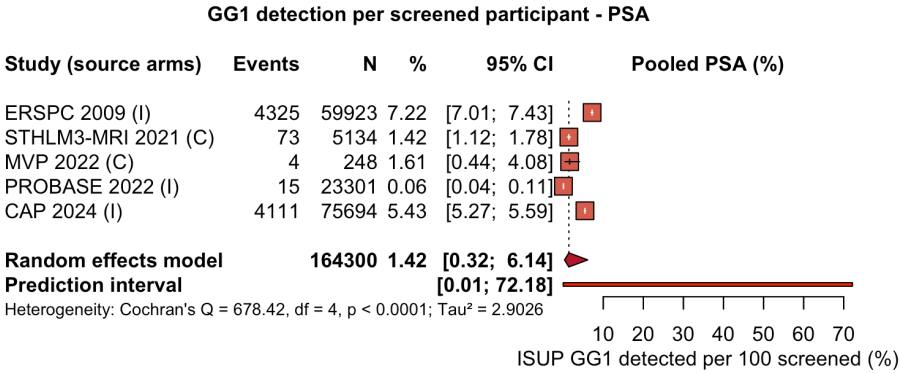


Figure 8c. Forest plot for GG1 per screened participant — PSA.

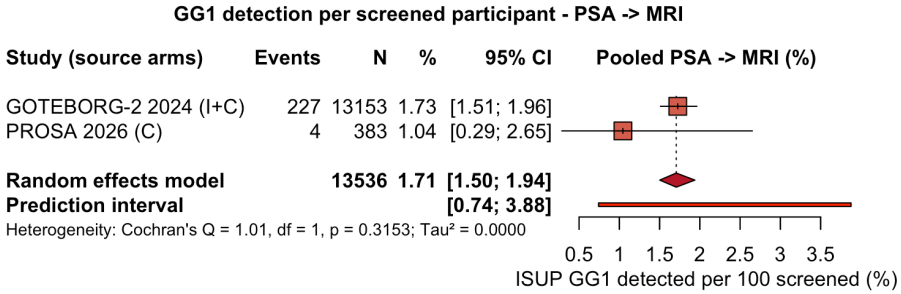


Figure 8d. Forest plot for GG1 per screened participant — PSA → MRI.

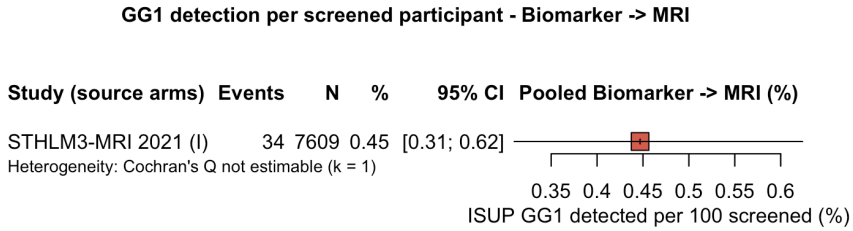


Figure 8e. Forest plot for GG1 per screened participant — biomarker → MRI.

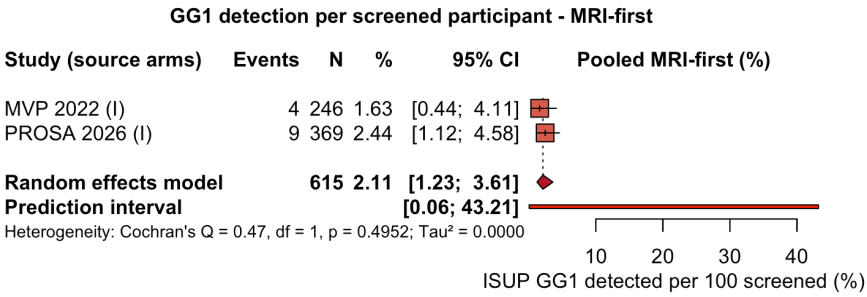


Figure 8f. Forest plot for GG1 per screened participant — MRI-first.

4.2 GG1 per randomized/invited participant

Strategy	k	Proportion (95% CI)	I²
No screening / usual care	3	2.702% (1.634–4.438)	99.9%
PSA + DRE	1	5.918% (5.684–6.159)	—
PSA	5	1.112% (0.287–4.216)	99.8%
PSA → MRI	2	1.706% (1.501–1.939)	2.9%
Biomarker → MRI	1	0.447% (0.310–0.624)	—
MRI-first	2	2.054% (1.196–3.504)	0.0%

The two denominators separate where screening uptake was incomplete. For PSA + DRE (PLCO) the rate is 6.962% per screened man against 5.918% per randomized man. Control arms of invitation trials were never screened, so their two denominators coincide, and the No screening / usual care rows in 4.1 and 4.2 are identical. The Biomarker → MRI rows also match, because STHLM3-MRI, its only contributor, reports the same screened and randomized denominator.

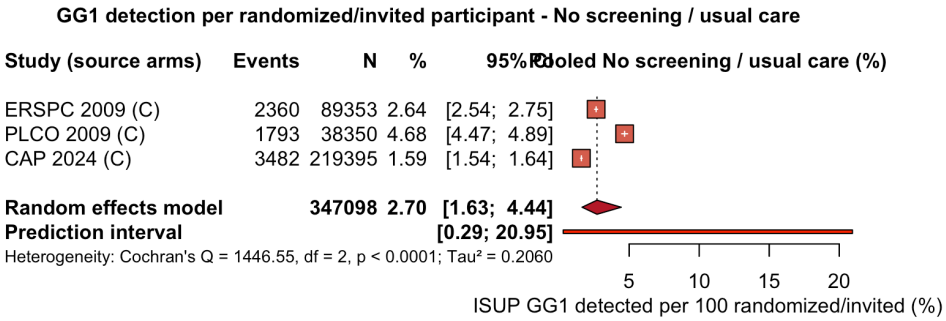


Figure 9a. Forest plot for GG1 per randomized/invited participant — no screening / usual care.

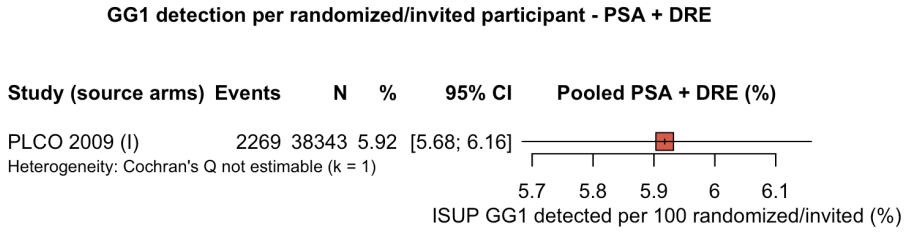


Figure 9b. Forest plot for GG1 per randomized/invited participant — PSA + DRE.

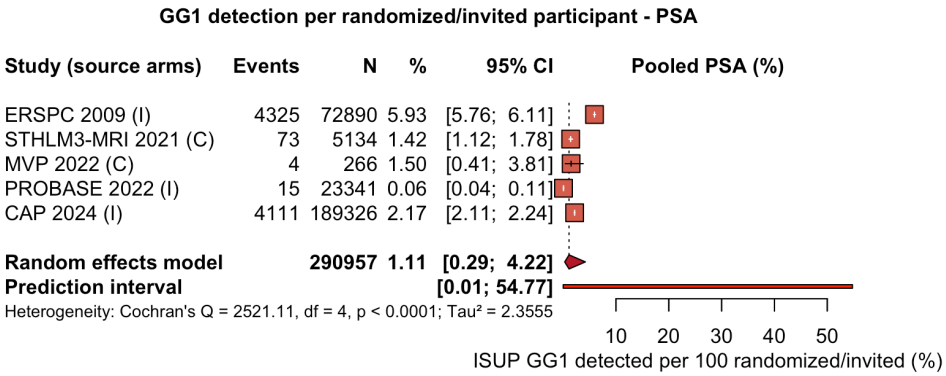


Figure 9c. Forest plot for GG1 per randomized/invited participant — PSA.

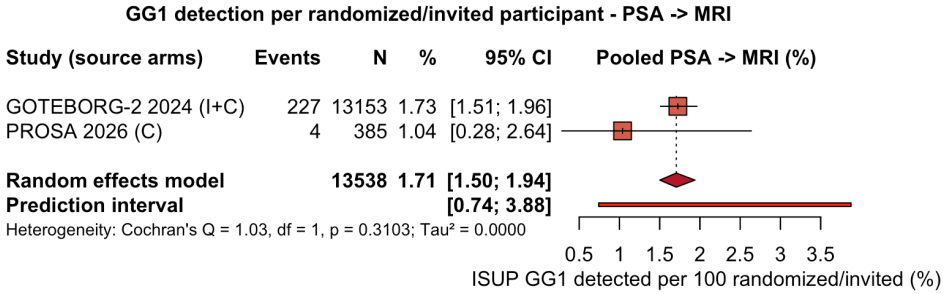


Figure 9d. Forest plot for GG1 per randomized/invited participant — PSA → MRI.

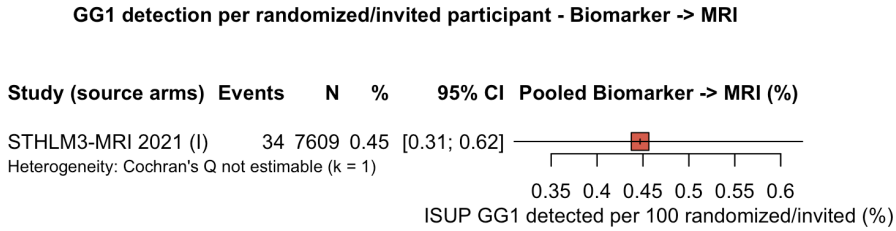


Figure 9e. Forest plot for GG1 per randomized/invited participant — biomarker → MRI.

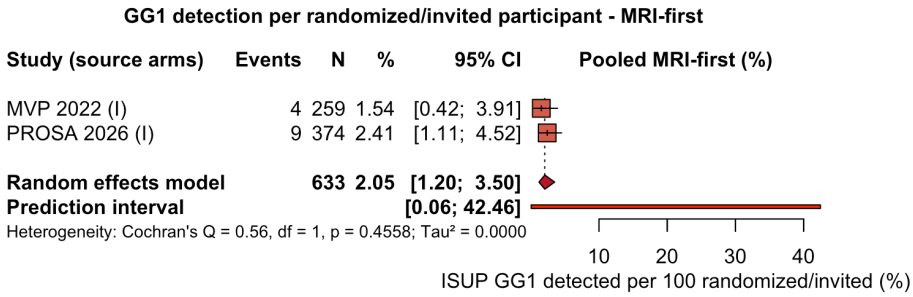


Figure 9f. Forest plot for GG1 per randomized/invited participant — MRI-first.

4.3 GG1 among all detected prostate cancers

Strategy	k	Proportion (95% CI)	I²
No screening / usual care	3	46.730% (30.240–63.967)	99.9%
PSA + DRE	1	65.730% (64.120–67.314)	—
PSA	5	45.788% (31.125–61.218)	99.8%
PSA → MRI	2	46.761% (42.398–51.175)	0.0%
Biomarker → MRI	1	15.044% (10.649–20.383)	—
MRI-first	2	31.707% (19.387–47.266)	0.0%

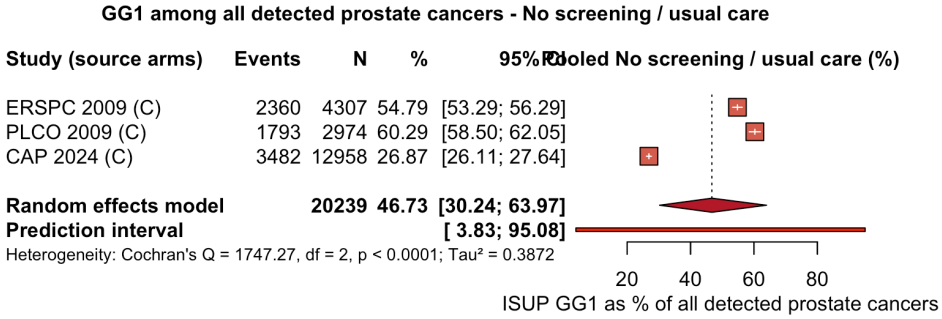


Figure 10a. Forest plot for GG1 among all detected prostate cancers — no screening / usual care.

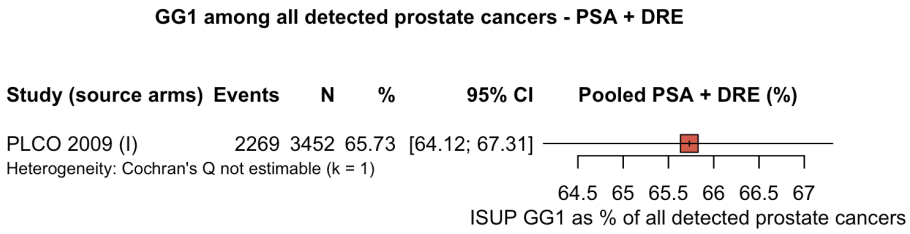


Figure 10b. Forest plot for GG1 among all detected prostate cancers — PSA + DRE.

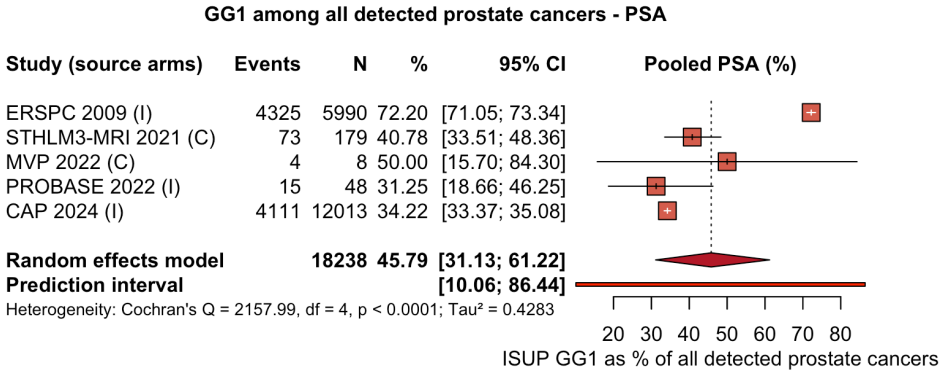


Figure 10c. Forest plot for GG1 among all detected prostate cancers — PSA.

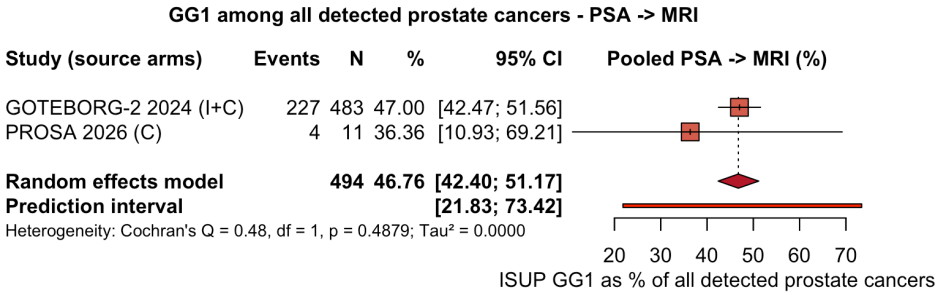


Figure 10d. Forest plot for GG1 among all detected prostate cancers — PSA → MRI.

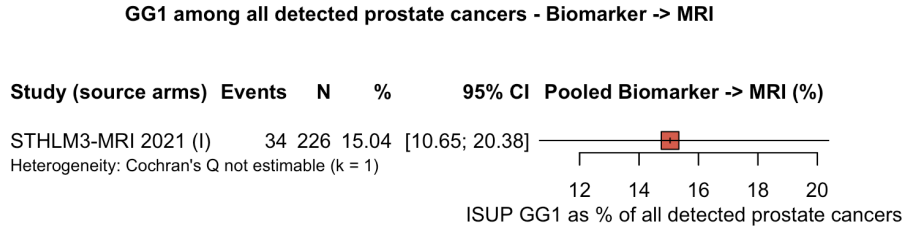


Figure 10e. Forest plot for GG1 among all detected prostate cancers — biomarker → MRI.

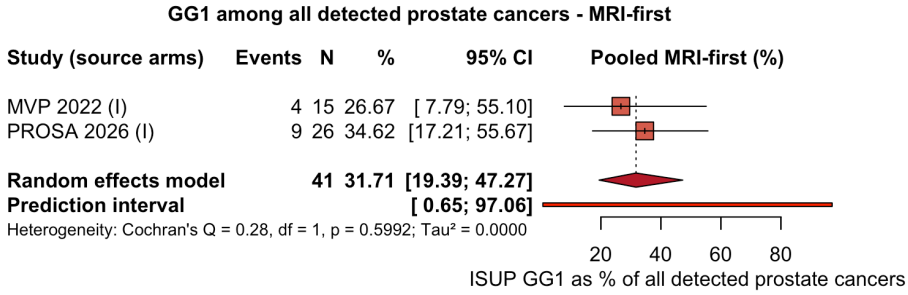


Figure 10f. Forest plot for GG1 among all detected prostate cancers — MRI-first.

4.4 GG≥2 per screened participant

Strategy	k	Proportion (95% CI)	I²
No screening / usual care	3	2.858% (2.245–3.631)	99.6%
PSA + DRE	1	3.387% (3.194–3.590)	—
PSA	5	1.629% (0.478–5.398)	99.8%
PSA → MRI	2	1.943% (1.724–2.190)	0.0%
Biomarker → MRI	1	2.523% (2.183–2.901)	—
MRI-first	2	4.553% (3.162–6.515)	0.0%

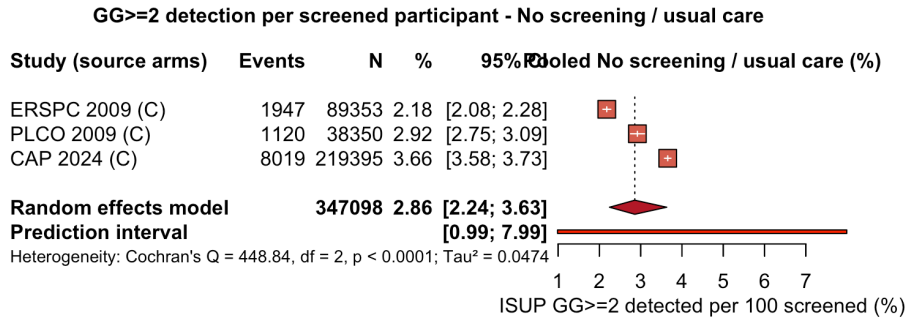


Figure 11a. Forest plot for GG≥2 per screened participant — no screening / usual care.

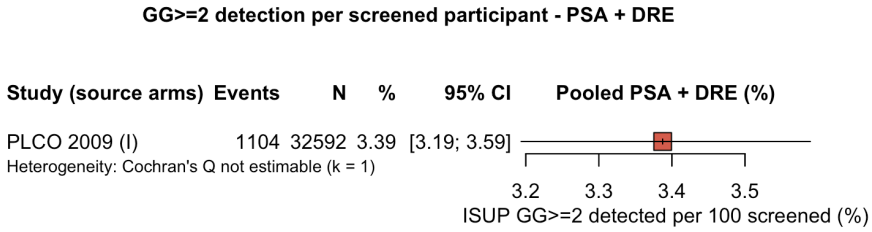


Figure 11b. Forest plot for GG≥2 per screened participant — PSA + DRE.

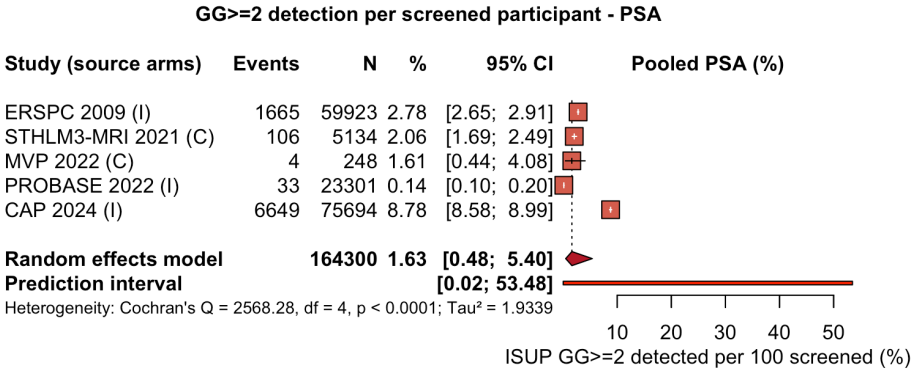


Figure 11c. Forest plot for GG≥2 per screened participant — PSA.

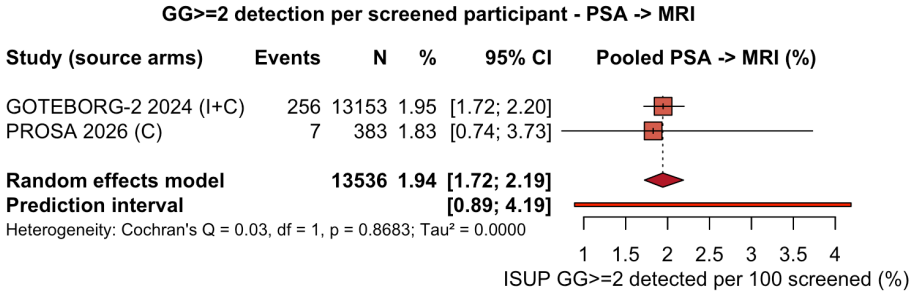


Figure 11d. Forest plot for GG≥2 per screened participant — PSA → MRI.

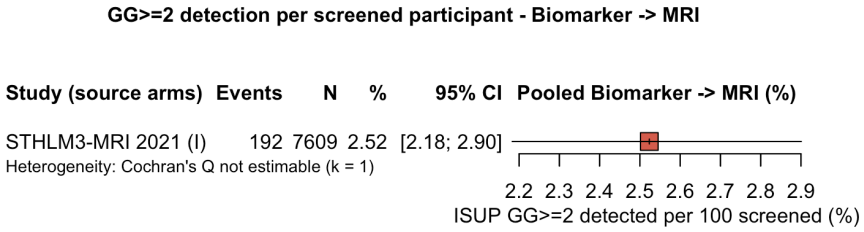


Figure 11e. Forest plot for GG≥2 per screened participant — biomarker → MRI.

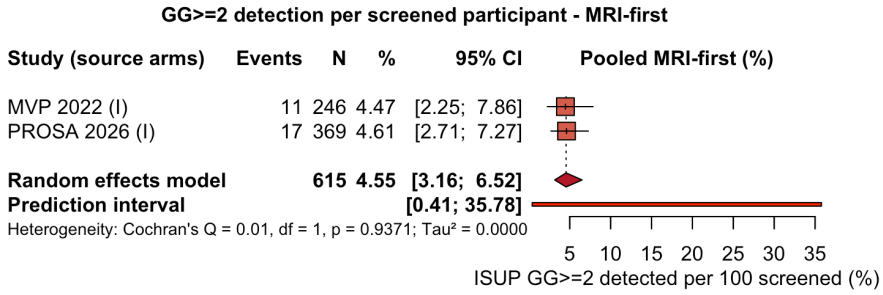


Figure 11f. Forest plot for GG≥2 per screened participant — MRI-first.

4.5 GG≥2 per randomized/invited participant

Strategy	k	Proportion (95% CI)	I²
No screening / usual care	3	2.858% (2.245–3.631)	99.6%
PSA + DRE	1	2.879% (2.714–3.052)	—
PSA	5	1.271% (0.456–3.491)	99.3%
PSA → MRI	2	1.943% (1.723–2.189)	0.0%
Biomarker → MRI	1	2.523% (2.183–2.901)	—
MRI-first	2	4.423% (3.071–6.332)	0.0%

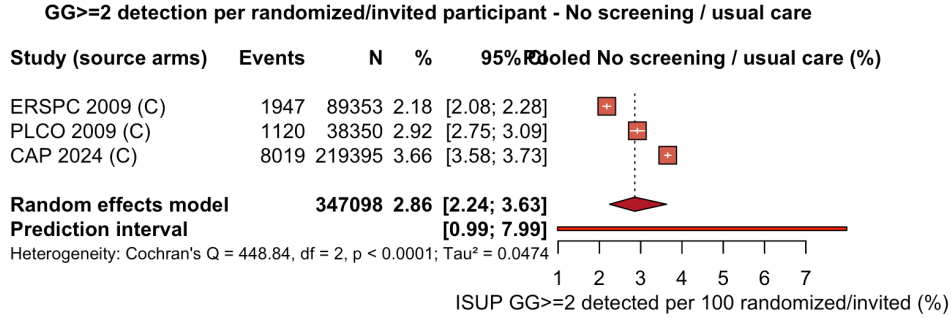


Figure 12a. Forest plot for GG≥2 per randomized/invited participant — no screening / usual care.

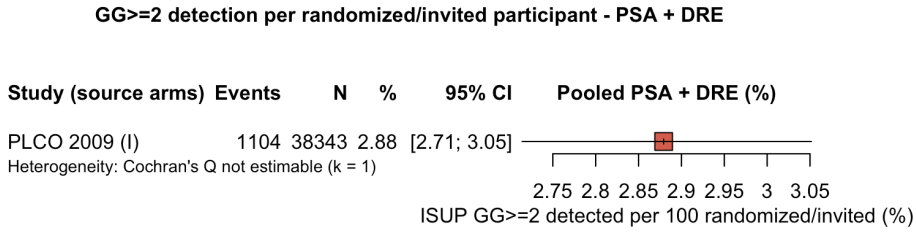


Figure 12b. Forest plot for GG≥2 per randomized/invited participant — PSA + DRE.

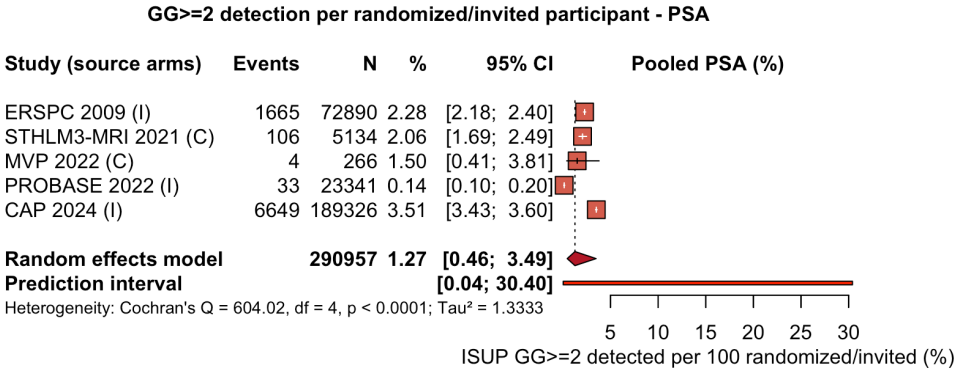


Figure 12c. Forest plot for GG≥2 per randomized/invited participant — PSA.

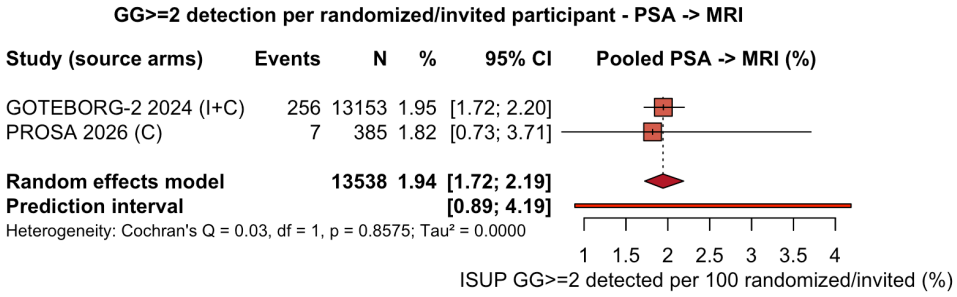


Figure 12d. Forest plot for GG≥2 per randomized/invited participant — PSA → MRI.

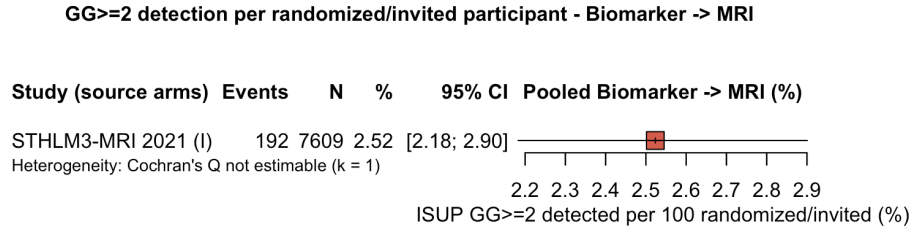


Figure 12e. Forest plot for GG≥2 per randomized/invited participant — biomarker → MRI.

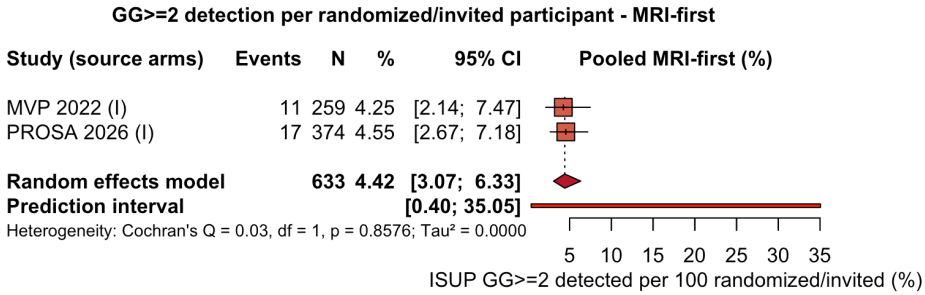


Figure 12f. Forest plot for GG≥2 per randomized/invited participant — MRI-first.

4.6 GG≥2 among all detected prostate cancers

Strategy	k	Proportion (95% CI)	I²
No screening / usual care	3	48.262% (36.887–59.820)	99.7%
PSA + DRE	1	31.982% (30.427–33.567)	—
PSA	5	51.494% (37.320–65.432)	99.7%
PSA → MRI	2	53.239% (48.825–57.602)	0.0%
Biomarker → MRI	1	84.956% (79.617–89.351)	—
MRI-first	2	68.293% (52.734–80.613)	0.0%

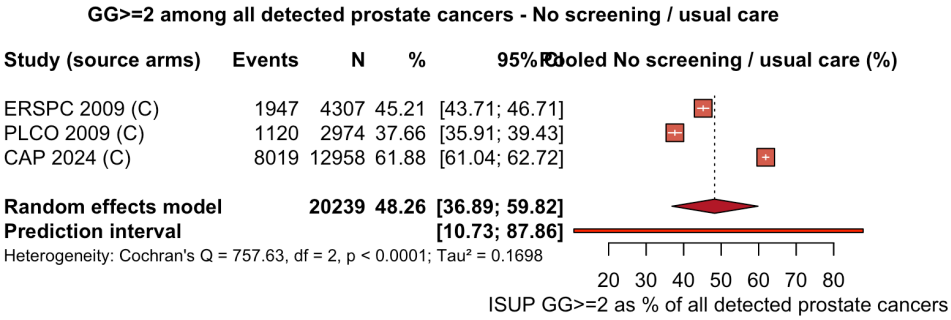


Figure 13a. Forest plot for GG≥2 among all detected prostate cancers — no screening / usual care.

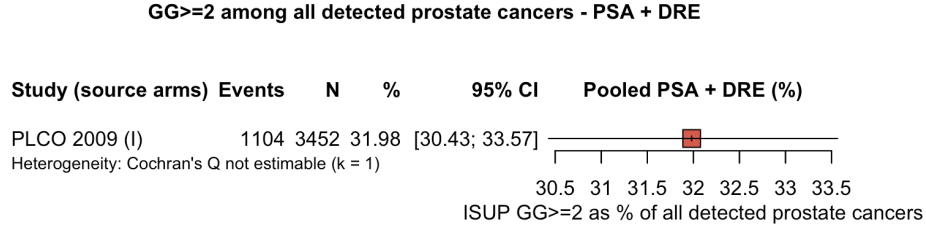


Figure 13b. Forest plot for GG≥2 among all detected prostate cancers — PSA + DRE.

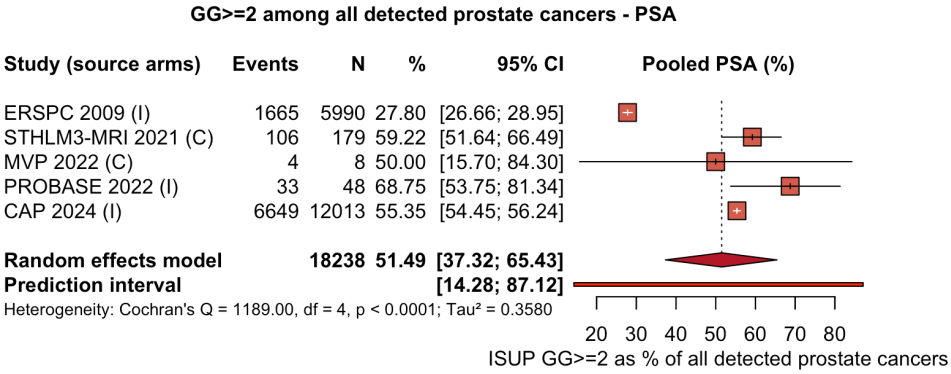


Figure 13c. Forest plot for GG≥2 among all detected prostate cancers — PSA.

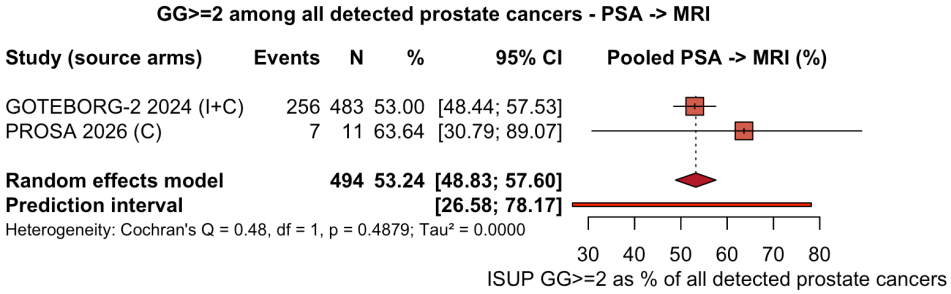


Figure 13d. Forest plot for GG≥2 among all detected prostate cancers — PSA → MRI.

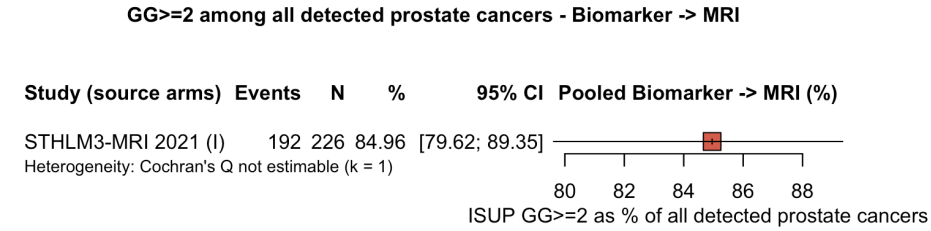


Figure 13e. Forest plot for GG≥2 among all detected prostate cancers — biomarker → MRI.

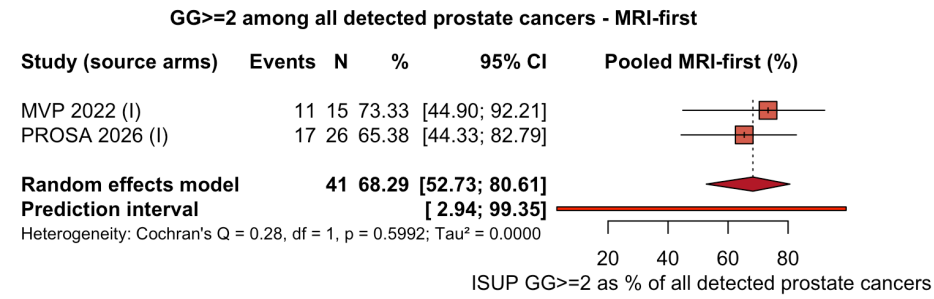


Figure 13f. Forest plot for GG≥2 among all detected prostate cancers — MRI-first.

4.7 GG1 among participants biopsied

Strategy	k	Proportion (95% CI)	I²
PSA	3	18.164% (11.539–27.415)	81.0%
PSA → MRI	2	26.798% (23.947–29.856)	0.0%
MRI-first	2	20.312% (12.177–31.910)	0.0%

STHLM3-MRI was excluded from this outcome-specific analysis, so no Biomarker → MRI unit remains.

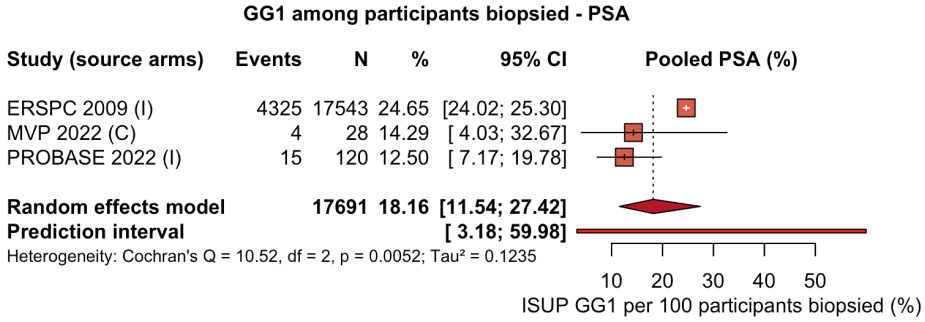


Figure 14a. Forest plot for GG1 among participants biopsied — PSA.

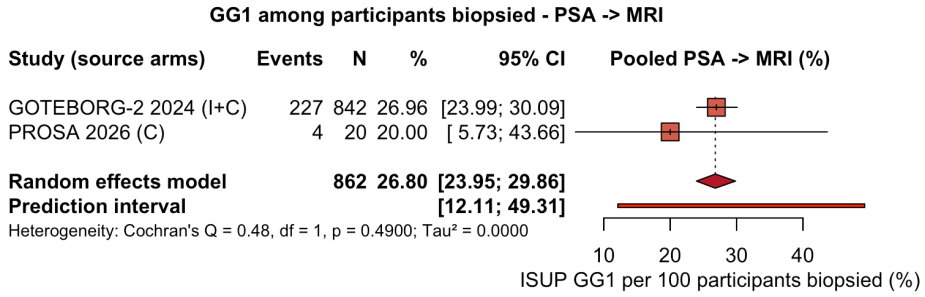


Figure 14b. Forest plot for GG1 among participants biopsied — PSA → MRI.

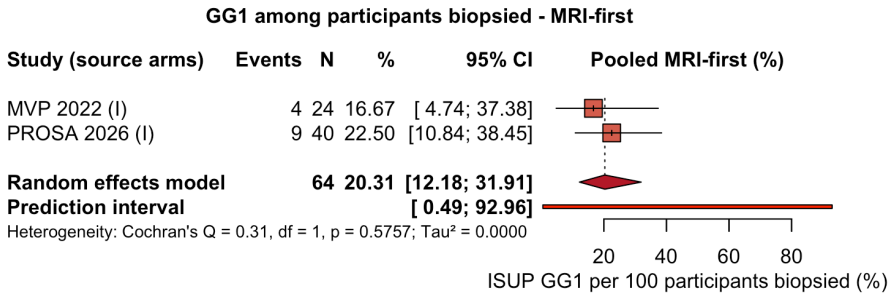


Figure 14c. Forest plot for GG1 among participants biopsied — MRI-first.

4.8 GG≥2 among participants biopsied

Strategy	k	Proportion (95% CI)	I²
PSA	3	15.471% (8.370–26.834)	94.9%
PSA → MRI	2	30.510% (27.526–33.668)	0.0%
MRI-first	2	43.750% (32.187–56.034)	0.0%

STHLM3-MRI was excluded from this outcome-specific analysis, so no Biomarker → MRI unit remains.

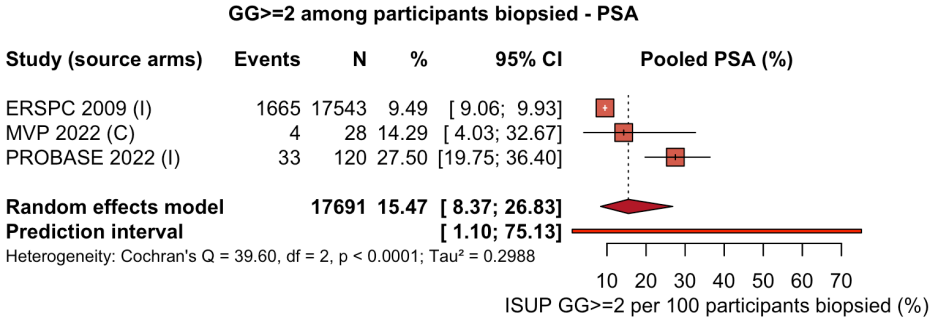


Figure 15a. Forest plot for GG≥2 among participants biopsied — PSA.

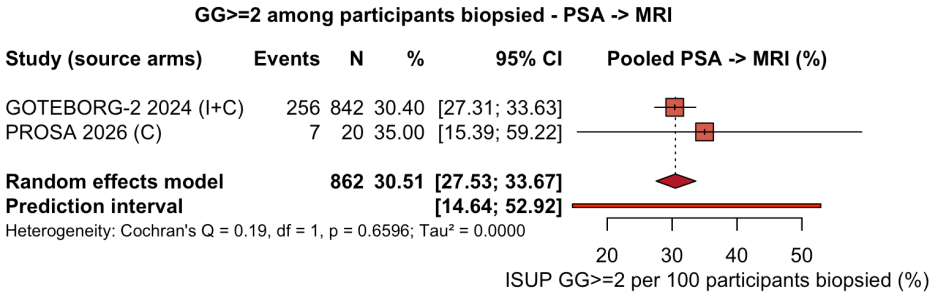


Figure 15b. Forest plot for GG≥2 among participants biopsied — PSA → MRI.

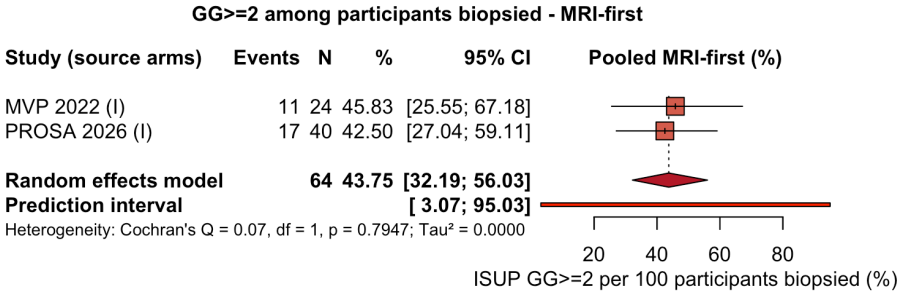


Figure 15c. Forest plot for GG≥2 among participants biopsied — MRI-first.

4.9 Biopsies per screened/assessed participant

Strategy	k	Proportion (95% CI)	I²
PSA	5	6.811% (2.018–20.599)	100.0%
PSA → MRI	2	6.368% (5.969–6.792)	0.0%
Biomarker → MRI	1	4.087% (3.654–4.557)	—
MRI-first	2	10.406% (8.229–13.079)	0.0%

This outcome uses screened/assessed rather than randomized/invited denominators. For CAP, the calculation is 5,848/75,694 (7.7%), not 5,848/189,326 (3.1%).

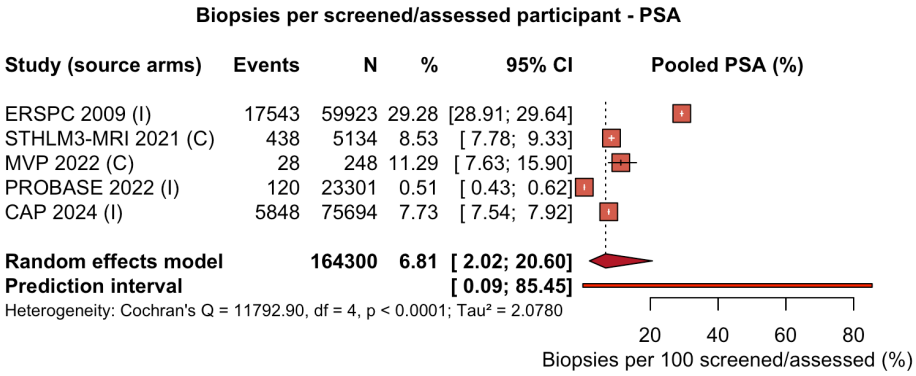


Figure 16a. Forest plot for Biopsies per screened/assessed participant — PSA.

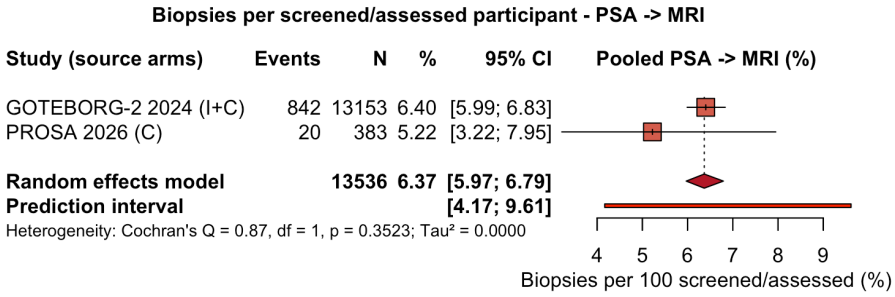


Figure 16b. Forest plot for Biopsies per screened/assessed participant — PSA → MRI.

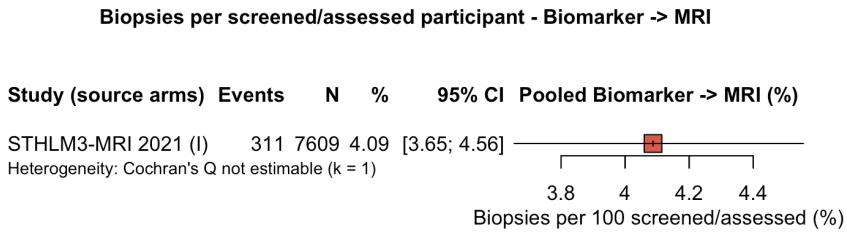


Figure 16c. Forest plot for Biopsies per screened/assessed participant — biomarker → MRI.

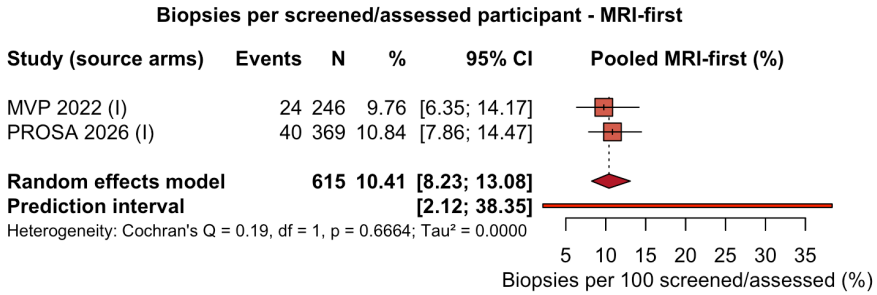


Figure 16d. Forest plot for Biopsies per screened/assessed participant — MRI-first.

5. Sensitivity analyses

Analysis	Outcome	k	Estimate (95% CI)
Invitation cohort excluding CAP	GG1 / randomized	2	RR 1.687 (0.044–64.583)
Invitation cohort excluding CAP	GG1 / all cancers	2	RR 1.199 (0.360–3.996)
Invitation cohort excluding CAP	GG≥2 / randomized	2	RR 1.022 (0.696–1.501)
Invitation cohort excluding CAP	GG≥2 / all cancers	2	RR 0.722 (0.093–5.616)
Invitation cohort excluding ERSPC	GG1 / randomized	2	RR 1.320 (0.806–2.161)
Invitation cohort excluding ERSPC	GG1 / all cancers	2	RR 1.178 (0.439–3.162)
Invitation cohort excluding ERSPC	GG≥2 / randomized	2	RR 0.964 (0.863–1.077)
Invitation cohort excluding ERSPC	GG≥2 / all cancers	2	RR 0.880 (0.651–1.191)
MVP/PROSA assessed denominators	GG1 detection per assessed participant	4	RR 0.562 (0.152–2.083)
MVP/PROSA assessed denominators	GG≥2 detection per assessed participant	4	RR 1.291 (0.572–2.913)

PROBASE standalone GG≥2 continuity-correction sensitivity:

Added value	RR (95% CI)
0.1	330.433 (0.666–164009.952)
0.5	66.885 (4.098–1091.529)
1.0	33.942 (4.647–247.929)

6. Additional checks and limitations

The analysis generated 97 leave-one-out refits. Complete results are in output/table3_leave_one_out.csv. Analyses with fewer than three contributing studies have no leave-one-out rows by design.

Publication-bias analysis was not performed because the largest model contains five studies, below the protocol threshold of ten. The threshold is evaluated at run time, so a future search update that pushes any model to ten or more studies triggers Peters' regression and funnel plots automatically.

Protocol deviations

Recorded in full at the end of output/results.txt.

1. **Stratification of comparative analyses (protocol s.8).** The protocol specifies pooling risk ratios "stratified by the screening strategy". A risk ratio is a contrast *between* two strategies, so a single strategy label cannot describe one. Trials are grouped instead by contrast family (invitation, diagnostic/biopsy strategy, timing), and families are never pooled together. Single-arm proportions are stratified by screening strategy exactly as the protocol states.
2. **Continuity correction for proportions (protocol s.8).** The protocol states a 0.5 correction for zero cells in both analyses. It is applied to the risk-ratio models. The proportion models are logit-normal GLMMs fitted by likelihood; they use the exact binomial likelihood and need no continuity correction, and adding one would bias them. None is applied there.
3. **Curative-treatment outcome (protocol s.2 and s.9).** "Percentage of patients receiving curative treatment over the course of follow-up" is a registered secondary outcome. The included reports provide no extractable arm-level treatment counts, so it is not analyzed. This is an outcome-level deviation.
4. **Adverse events and procedural harms (protocol s.9).** Contextual data only, not formal endpoints; not extracted and not summarized.
5. **Exclusion of an extracted trial.** ProScreen was extracted (its rows remain in `arm_level_data.csv`) but is excluded from every analysis, including the descriptive tables.

RoB 2 and the PRISMA flow diagram are handled outside this pipeline.

7. Output files

- output/table1_descriptives.csv: arm-level data.
- output/table1b_study_strategy_units.csv: strategy-proportion inputs.
- output/table2_pooled_estimates.csv: pooled estimates, heterogeneity, and prediction intervals.
- output/table3_leave_one_out.csv: leave-one-out results.
- `output/fig1_*` – `output/fig7_*`: comparative and timing RR forest plots.
- `output/fig8_*` – `output/fig16_*`: strategy-proportion forest plots.

Reproducibility checks: validate_analysis.R verifies the structure of the generated outputs, and tests/test_calculations.R unit-tests the calculation helpers against hand-computed values and independent `metafor` refits.